

TAIAROA HEAD, New Zealand

WMO: 938960

Lat: 45.775S

Lon: 170.727E

Elev: 76

StdP: 100.42

Time Zone: 12.00 (NZL)

Period: 94-02

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.7	3.5	-4.0	2.7	7.4	-2.7	3.0	6.8	23.2	6.2	21.5	5.3	10.2	240	0.805

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.5	20.8	14.5	19.2	13.8	17.9	13.3	16.2	18.6	15.4	17.5	14.8	16.7	7.9	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	10.9	17.1	14.6	10.4	16.3	14.0	10.0	15.6	45.8	18.9	43.4	17.5	41.8	16.8	20.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.5	20.9	18.3	DB	0.9	25.8	0.9	2.4	0.2	27.5	-0.3	28.9	-0.8	30.3	-1.5	32.0	(4)
			WB	-0.1	18.2	0.5	1.6	-0.5	19.3	-0.7	20.2	-1.0	21.1	-1.4	22.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	13.8	14.4	13.1	11.6	10.1	7.9	7.0	7.9	8.9	10.3	10.8	12.8
	DBStd	3.16	2.08	2.19	1.97	1.88	2.41	2.04	1.84	2.12	2.38	2.23	2.22	2.01
	HDD10.0	355	1	1	1	6	26	67	95	71	49	22	15	1
	HDD18.3	2787	142	113	162	201	254	312	351	322	283	250	226	173
	CDD10.0	613	118	123	98	56	31	6	3	7	16	31	39	87
	CDD18.3	3	1	1	1	0	0	0	0	0	0	0	0	1
	CDH23.3	12	1	4	3	0	0	0	0	0	0	0	0	4
	CDH26.7	1	0	0	0	0	0	0	0	0	0	0	0	1
Wind	WSAvg	7.4	7.0	6.5	7.3	7.3	7.4	7.4	7.0	7.6	7.4	7.8	8.1	7.4
Precipitation	PrecAvg	835	79	76	65	64	78	60	64	72	59	69	66	81
	PrecMax	1036	171	207	173	176	292	125	169	175	105	144	131	126
	PrecMin	641	20	23	19	23	20	26	25	19	22	24	26	35
	PrecStd	108	40	50	32	34	60	27	38	44	22	29	25	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.9	23.2	22.9	19.3	19.3	15.1	15.4	15.7	18.0	19.8	20.4	21.4
		MCWB	15.3	14.8	16.6	12.4	11.8	9.4	8.7	8.9	10.0	11.5	13.1	13.9
	2%	DB	19.6	20.5	19.8	17.2	17.2	13.7	12.2	14.1	15.9	17.1	17.4	18.6
		MCWB	14.9	15.1	14.0	11.8	10.9	8.7	7.4	8.4	9.3	10.6	11.5	13.1
	5%	DB	18.1	18.6	17.2	15.5	14.3	12.0	10.5	12.5	13.8	15.1	15.5	16.8
		MCWB	14.7	14.9	13.5	11.5	10.4	8.2	7.3	7.9	9.1	10.8	11.9	12.7
	10%	DB	16.7	17.2	15.8	14.2	12.9	10.8	9.5	10.8	12.3	13.4	14.0	15.6
		MCWB	14.3	14.5	13.1	11.4	10.0	7.9	7.0	7.4	8.7	10.2	11.2	12.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.1	17.5	16.7	14.0	12.6	10.2	9.3	10.0	11.2	13.1	14.5	15.7
		MCDB	19.4	20.2	21.7	16.5	15.8	12.8	11.6	13.0	15.1	16.9	17.5	17.6
	2%	WB	16.0	16.4	15.2	13.1	12.0	9.5	8.6	9.2	10.3	12.0	13.2	14.7
		MCDB	18.2	18.6	17.6	14.8	13.9	11.9	10.8	11.9	13.6	15.1	15.4	16.9
	5%	WB	15.3	15.6	14.4	12.4	11.4	9.0	8.1	8.6	9.7	11.3	12.4	14.0
		MCDB	17.2	17.3	16.2	14.1	13.4	11.3	9.7	11.1	12.6	13.9	14.3	16.0
	10%	WB	14.7	14.9	13.8	11.9	10.7	8.3	7.5	8.1	9.1	10.6	11.6	13.1
		MCDB	16.4	16.6	15.3	13.6	12.7	10.4	9.1	10.2	11.6	12.8	13.4	14.7
Mean Daily Temperature Range	5% DB	MDBR	4.3	4.5	4.4	4.0	3.9	3.5	3.3	3.9	4.3	4.4	4.6	4.5
		MCDBR	6.6	7.0	7.0	6.3	6.6	5.4	4.9	6.5	7.2	7.1	7.3	6.9
		MCWBR	4.2	3.6	3.8	3.7	3.4	3.3	3.2	3.5	3.8	3.8	4.5	4.4
	5% WB	MCDBR	5.5	5.4	5.8	4.7	4.5	4.5	3.5	4.6	5.8	5.5	5.4	5.8
		MCWBR	4.0	3.3	3.8	3.4	2.8	3.2	2.9	3.0	3.6	3.5	3.9	4.1
Clear-Sky Solar Irradiance	taub	0.332	0.330	0.312	0.299	0.287	0.282	0.278	0.297	0.310	0.319	0.320	0.328	
	taud	2.575	2.584	2.639	2.653	2.641	2.606	2.630	2.606	2.561	2.558	2.580	2.589	
	Ebn at Noon	988	961	932	870	802	760	803	857	917	963	995	999	
	Edh at Noon	102	95	81	67	55	51	54	68	86	96	100	102	
All-Sky Solar Radiation	RadAvg	5.82	5.17	3.85	2.48	1.51	1.15	1.41	2.12	3.36	4.71	5.74	6.15	
	RadStd	0.47	0.42	0.17	0.17	0.09	0.08	0.09	0.13	0.18	0.25	0.25	0.36	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	+0.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page